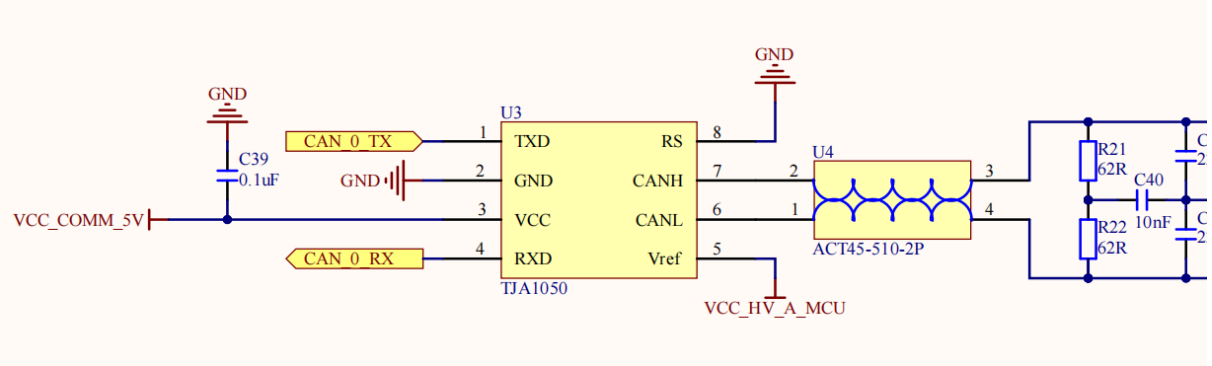
25、添加CanTrcv并解决PlatformType.h和Std_Types.h类型重定义问题

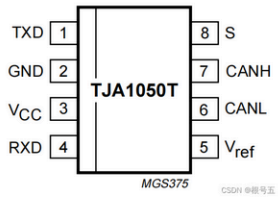
卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

(本文是为网络管理做准备，网络管理需要收发器！)

这里说明，我们板子使用的是TJA1050



TJA1050T芯片引脚

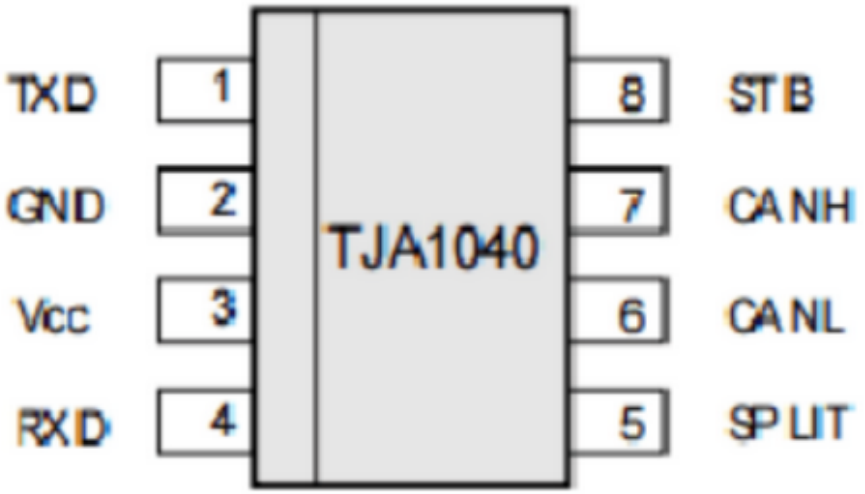


引脚说明如下：

引脚	名称	功能
1	TXD	**发送数据**，向CAN总线发送数据
2	GND	电源地
3	VCC	电源电压+5V
4	RXD	**接收数据**，从CAN总线读取数据
5	V_{ref}	参考电压输出
6	CANL	CAN总线**低**
7	CANH	CAN总线**高**
8	S	**低电平**：高速模式 **高电平**：静音模式

8脚直接接地，所有我们的CAN收发器一直是正常工作模式。

所有我们类比TJA1040 （不类比TJA1043因为其管脚多，与TJA1050管脚不类似），他们的管脚数量一样8Pin。



8脚功能差异，TJA1040 STB为低电平是正常工作，为高电平是低功耗模式，同时也可以监控总线信号，反馈到RX给MCU做唤醒

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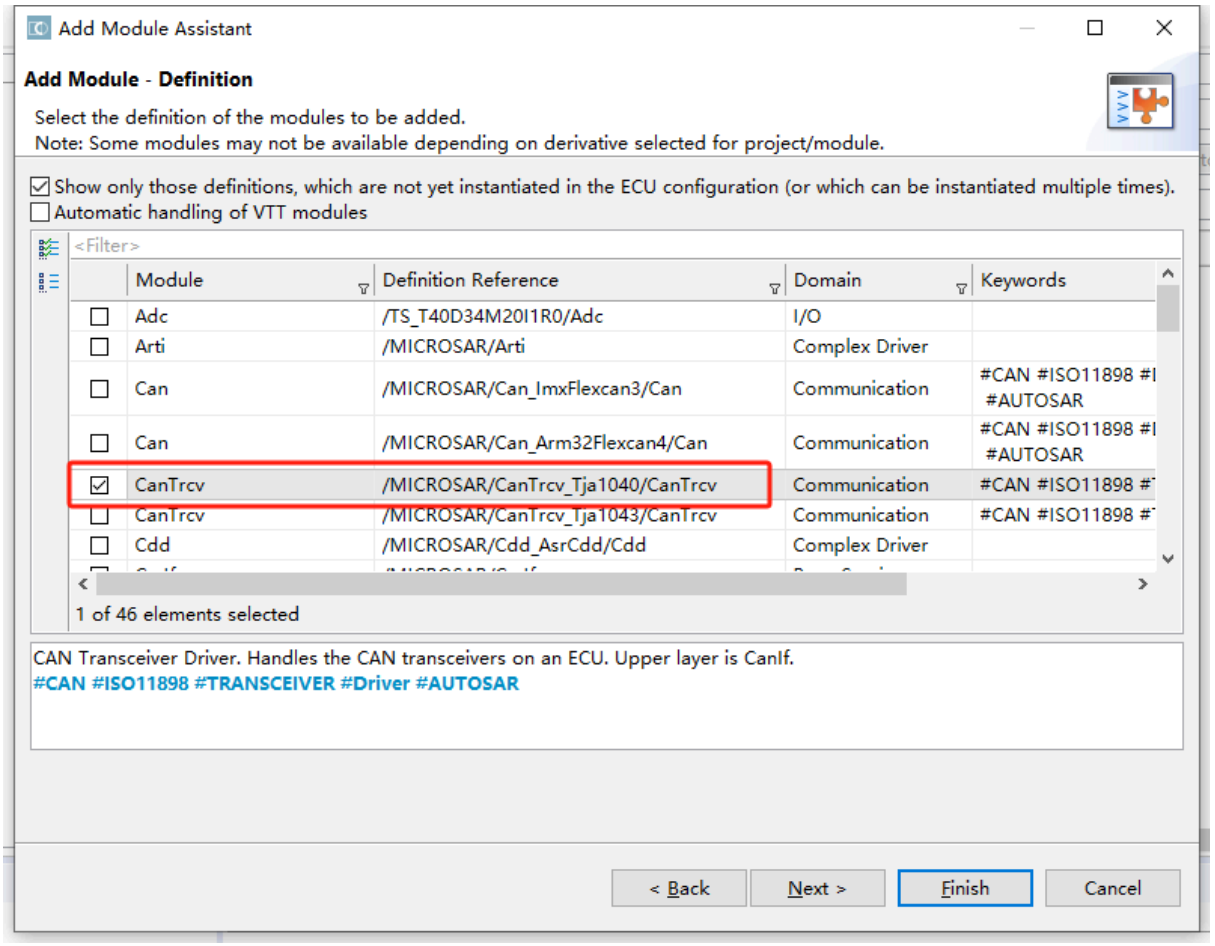
工作模式	工作模式特征	引脚 8 的电平	
		TJA1040	TJA1050
正常（高速）	发送功能 接收功能	低	低或不连接
待机	减少电流 远程唤醒 Babbling idiot 保护	高或不连接	————
静音	只接收 Babbling idiot 保护	-----	高

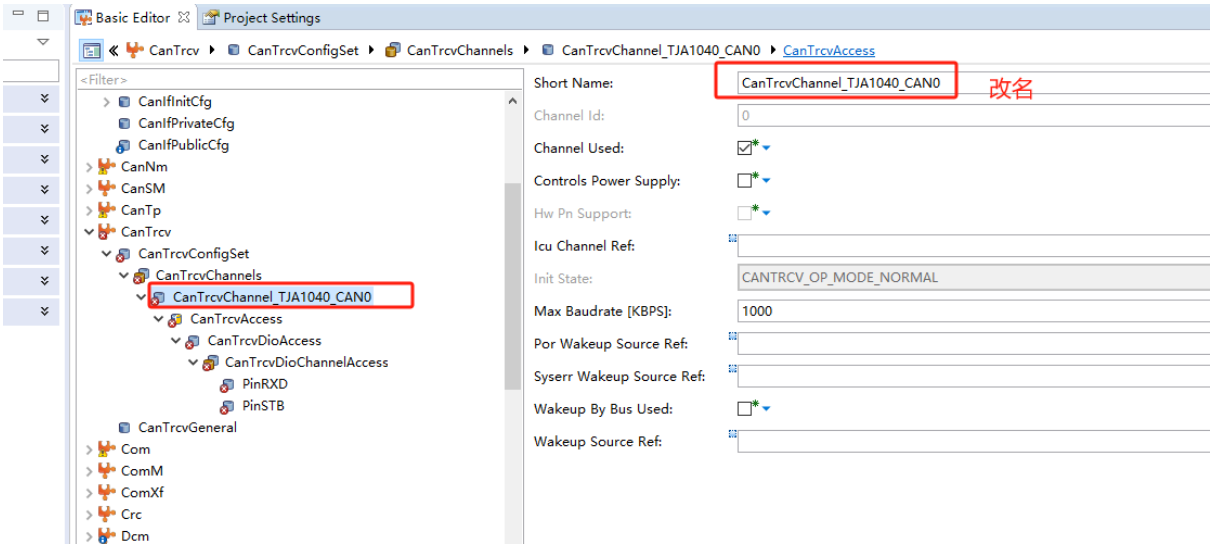
所以我们本文引用TJA1040作为替换例子， STB管脚我们映射为一个LED管脚，实际对我们功能不影响，因为我们只需要在Bus Sleep下通过报文发给收发器时 对应mcu rx管脚有电平变化，从而让收发器mainfunction可以检测到就能触发相关的代码。

一、添加CanTrcv Tja1040

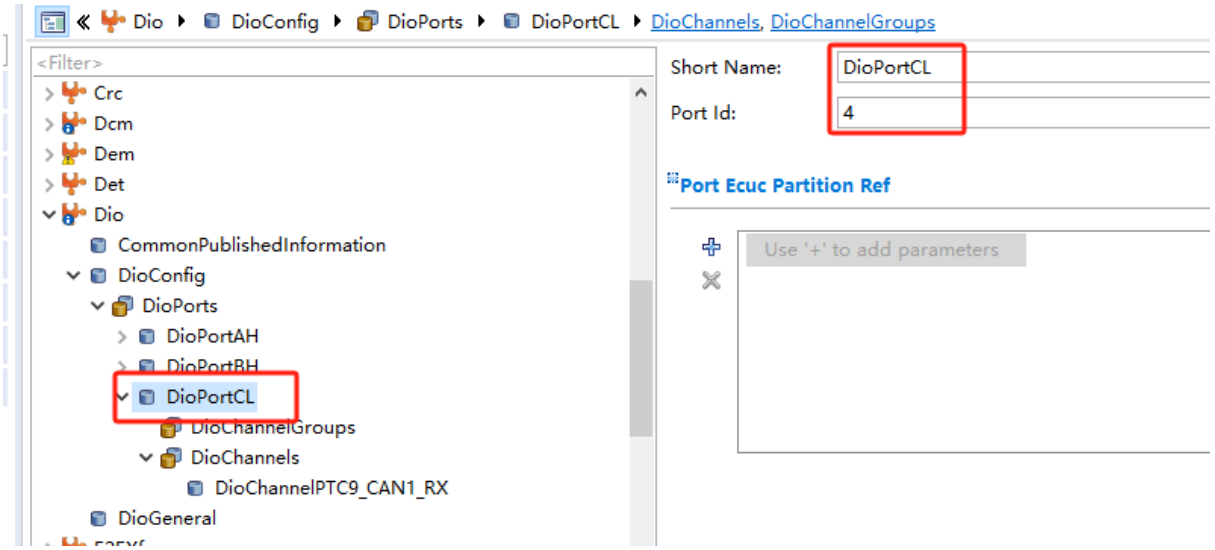
1.1、模块添加

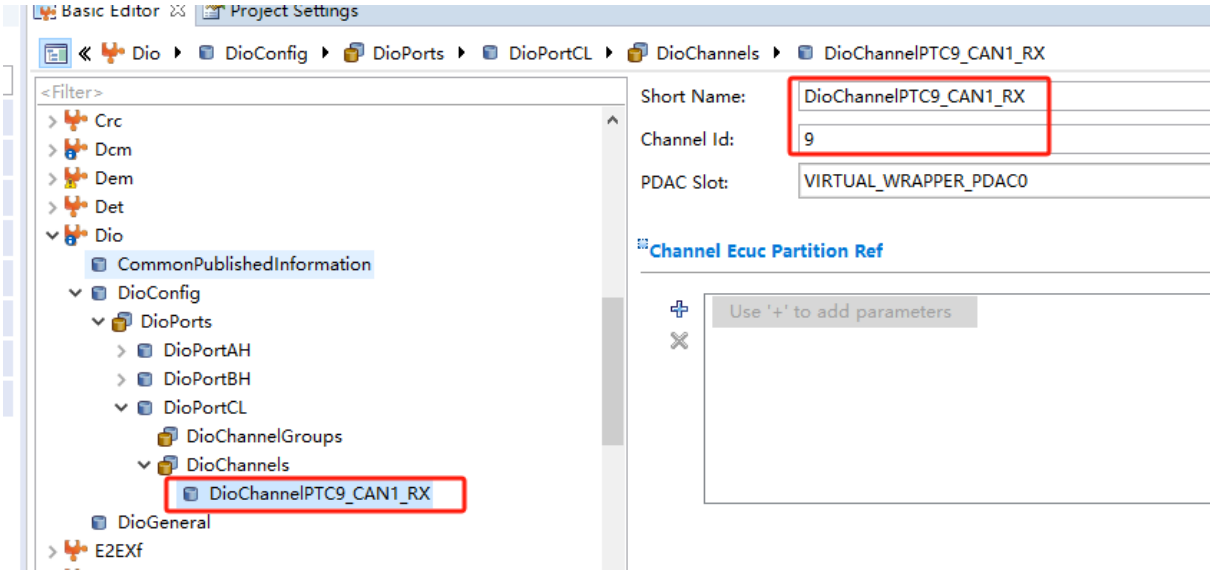


1.2、CanTrcv消错

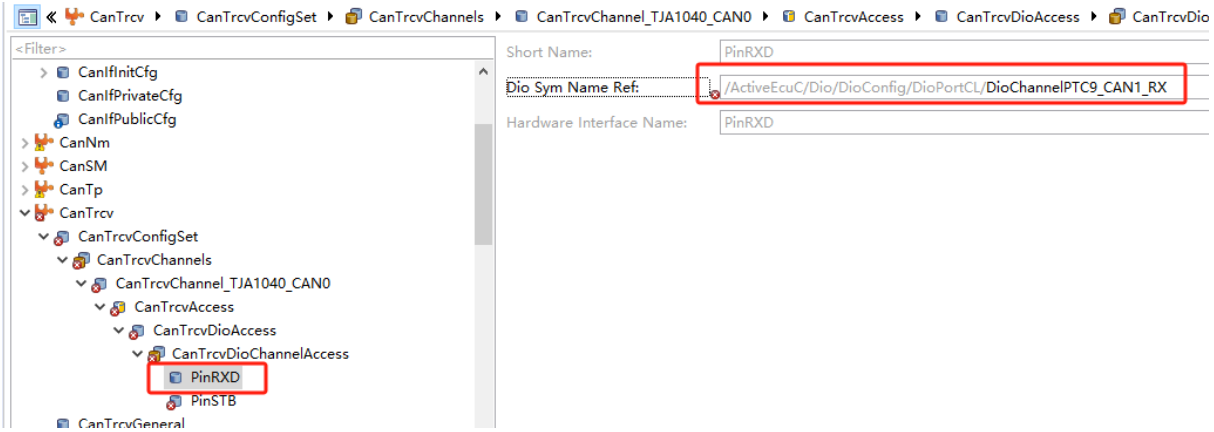


添加Can0 Rx 的dio

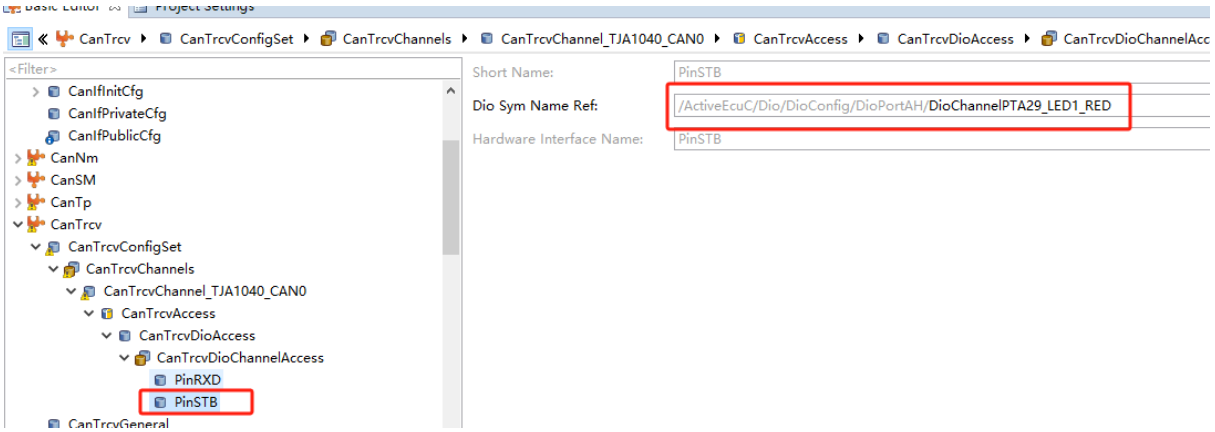




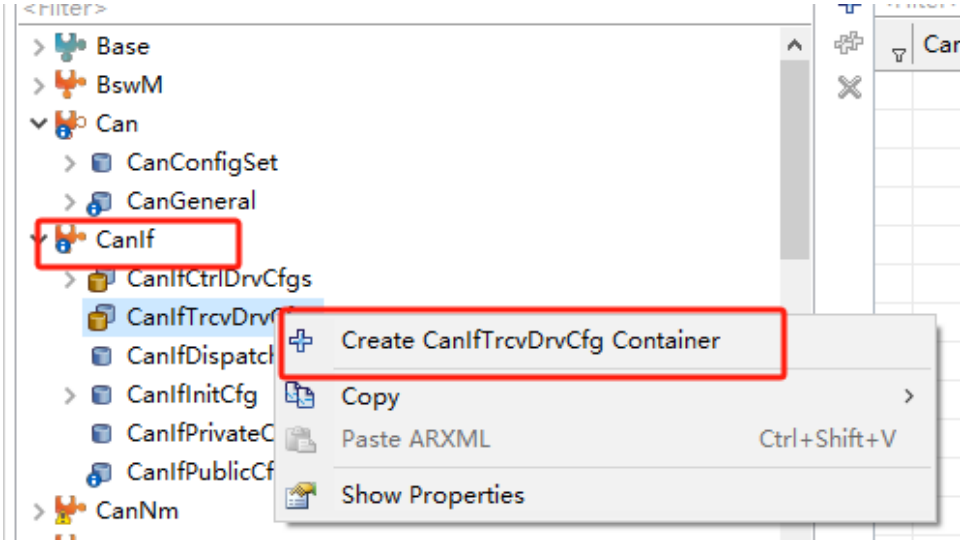
CanTrcv 的Rx选择Dio

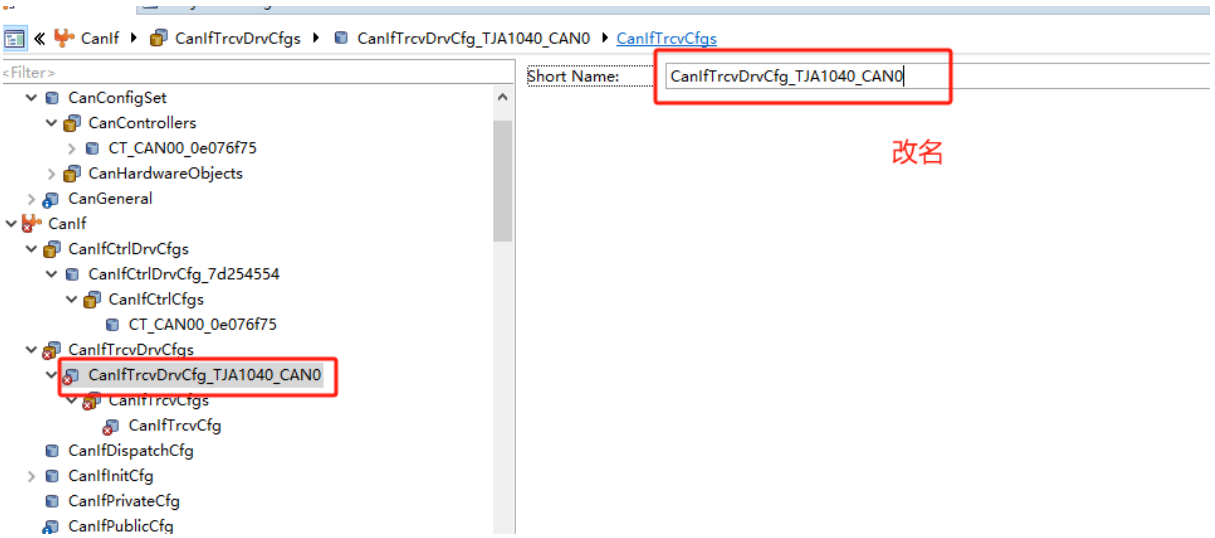


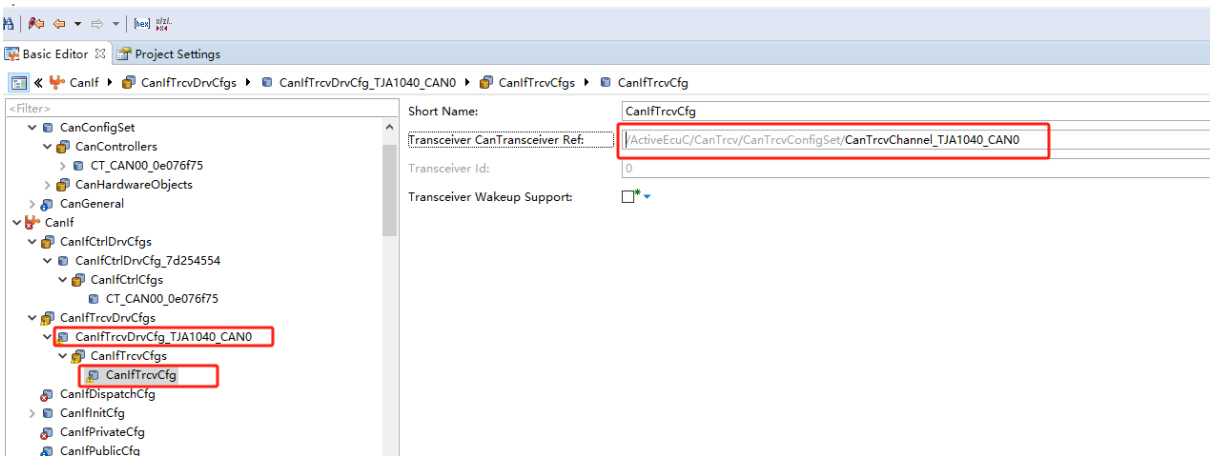
CanTrcv的STB 选择Dio，我们这里选择一个输出的Dio，LED1

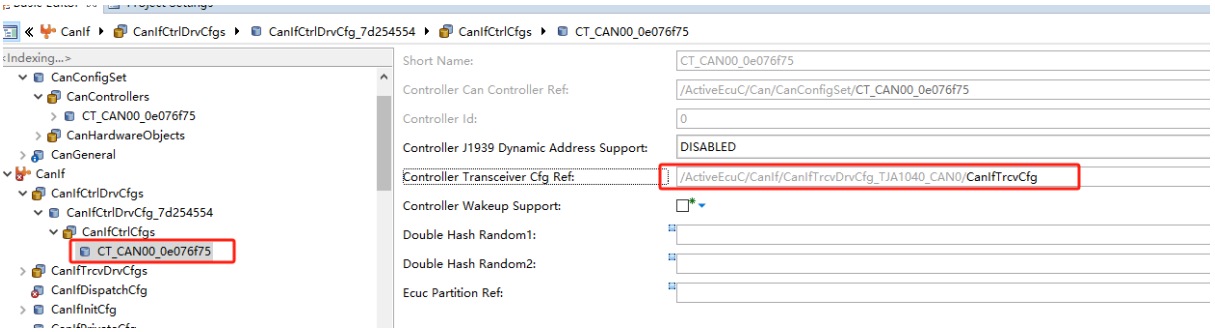


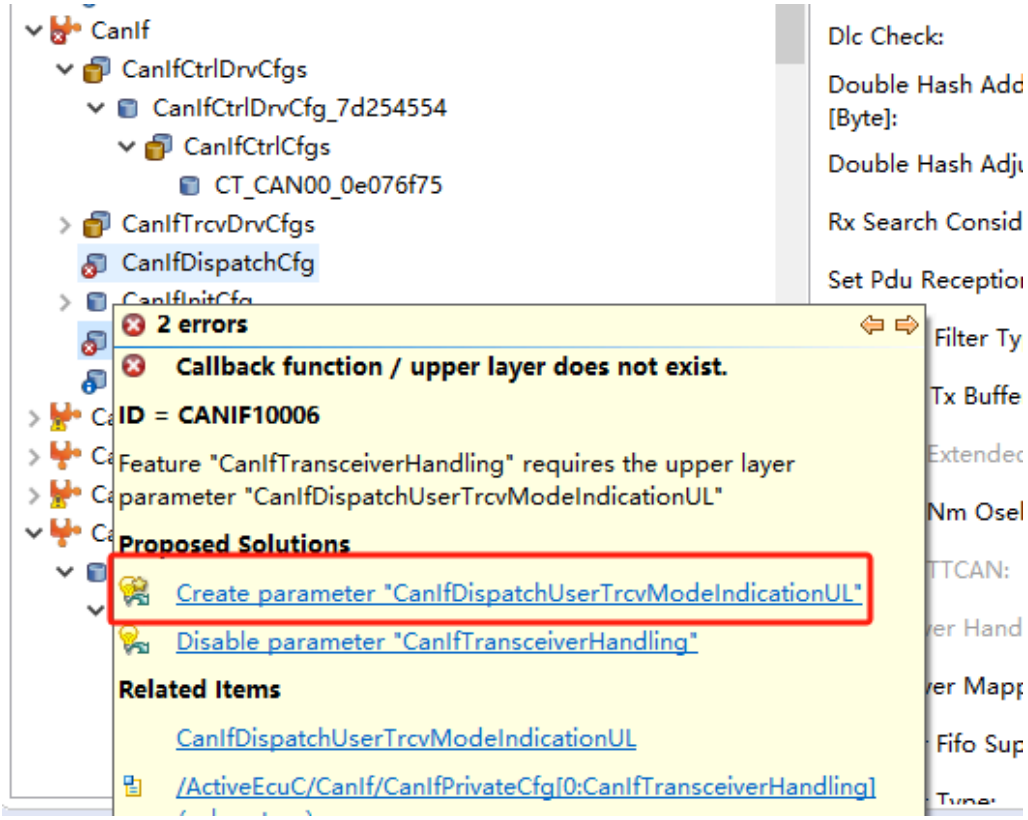
1.3、CanIf 添加CanTrcv的映射

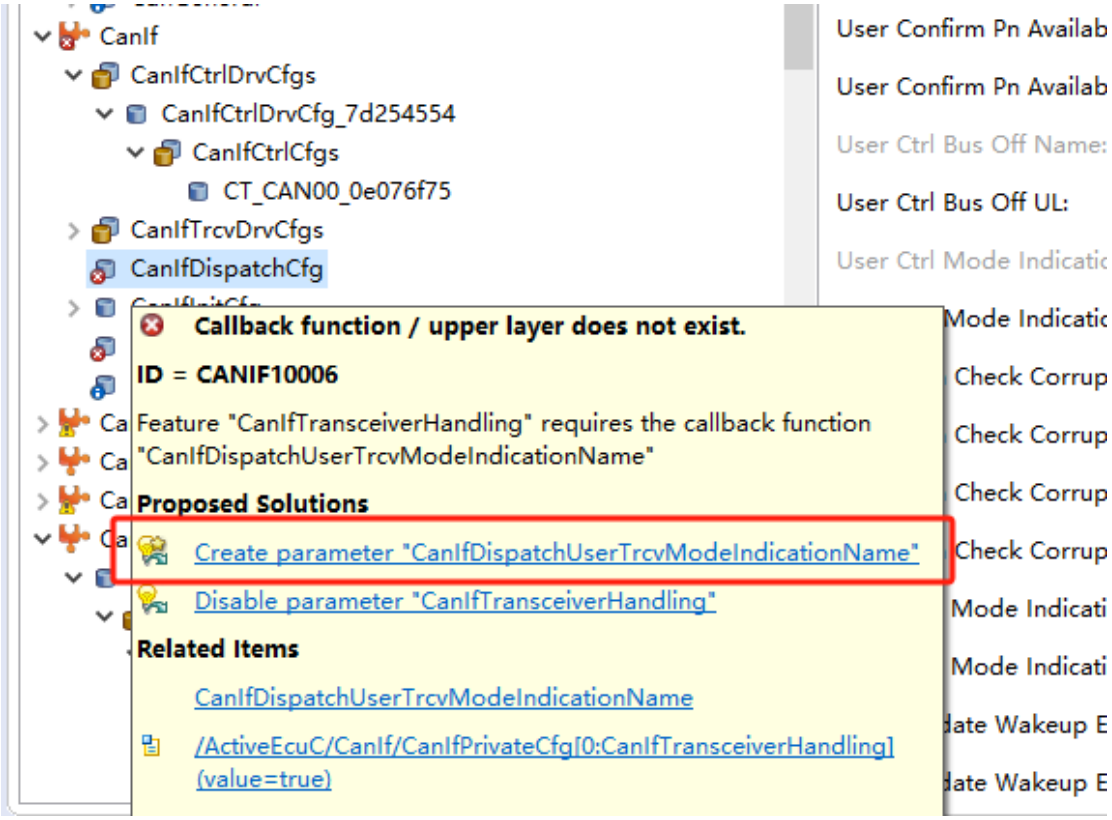




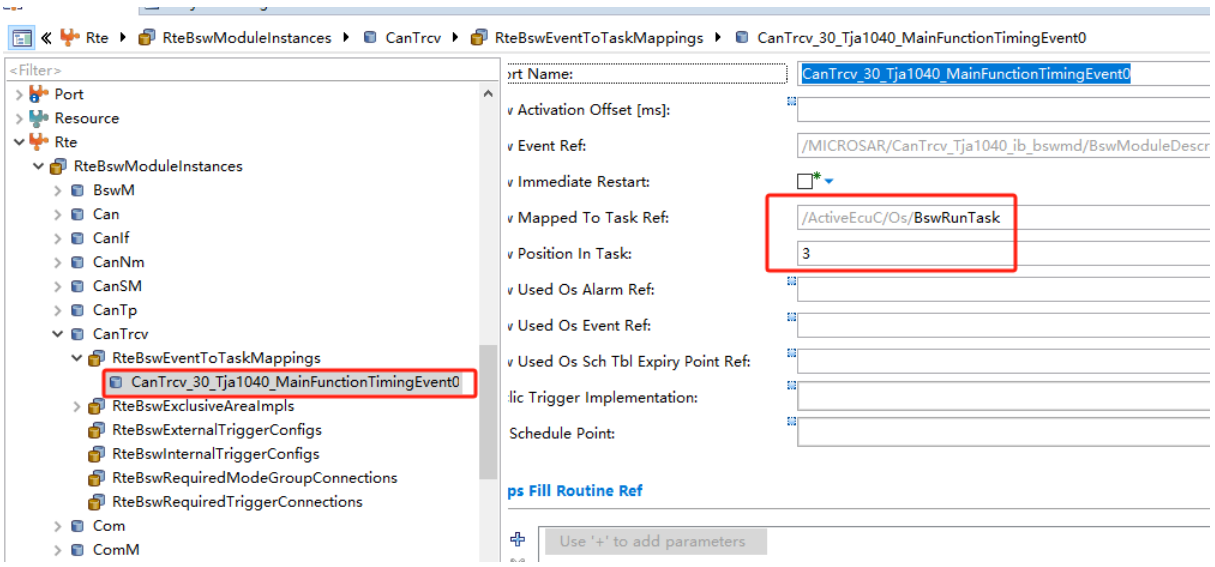

Basic Editor
Project Settings
CanIf ▸ CanIfTrcvDrvCfgs ▸ CanIfTrcvDrvCfg_TJA1040_CAN0 ▸ CanIfTrcvCfgs ▸ CanIfTrcvCfg
CanConfigSet
CanControllers
CT_CAN00_0e076f75
CanHardwareObjects
CanGeneral
CanIf
CanIfCtrlDrvCfgs
CanIfCtrlDrvCfg_7d254554
CanIfCtrlCfgs
CT_CAN00_0e076f75
CanIfTrcvDrvCfgs
CanIfTrcvDrvCfg_TJA1040_CAN0
CanIfTrcvCfgs
CanIfTrcvCfg
CanIfDispatchCfg
CanIfInitCfg
CanIfPrivateCfg
CanIfPublicCfg
Short Name:
CanIfTrcvCfg
Transceiver CanTransceiver Ref:
/ActiveEcuC/CanTrcv/CanTrcvConfigSet/CanTrcvChannel_TJA1040_CAN0
Transceiver Id:
0
Transceiver Wakeup Support:


CanIf ▸ CanIfCtrlDrvCfgs ▸ CanIfCtrlDrvCfg_7d254554 ▸ CanIfCtrlCfgs ▸ CT_CAN00_0e076f75
CanConfigSet
CanControllers
CT_CAN00_0e076f75
CanHardwareObjects
CanGeneral
CanIf
CanIfCtrlDrvCfgs
CanIfCtrlDrvCfg_7d254554
CanIfCtrlCfgs
CT_CAN00_0e076f75
CanIfTrcvDrvCfgs
CanIfDispatchCfg
CanIfInitCfg
Short Name:
CT_CAN00_0e076f75
Controller Can Controller Ref:
/ActiveEcuC/Can/CanConfigSet/CT_CAN00_0e076f75
Controller Id:
0
Controller J1939 Dynamic Address Support:
DISABLED
Controller Transceiver Cfg Ref:
/ActiveEcuC/CanIf/CanIfTrcvDrvCfg_TJA1040_CAN0/CanIfTrcvCfg
Controller Wakeup Support:
Double Hash Random1:
Double Hash Random2:
Ecuc Partition Ref:

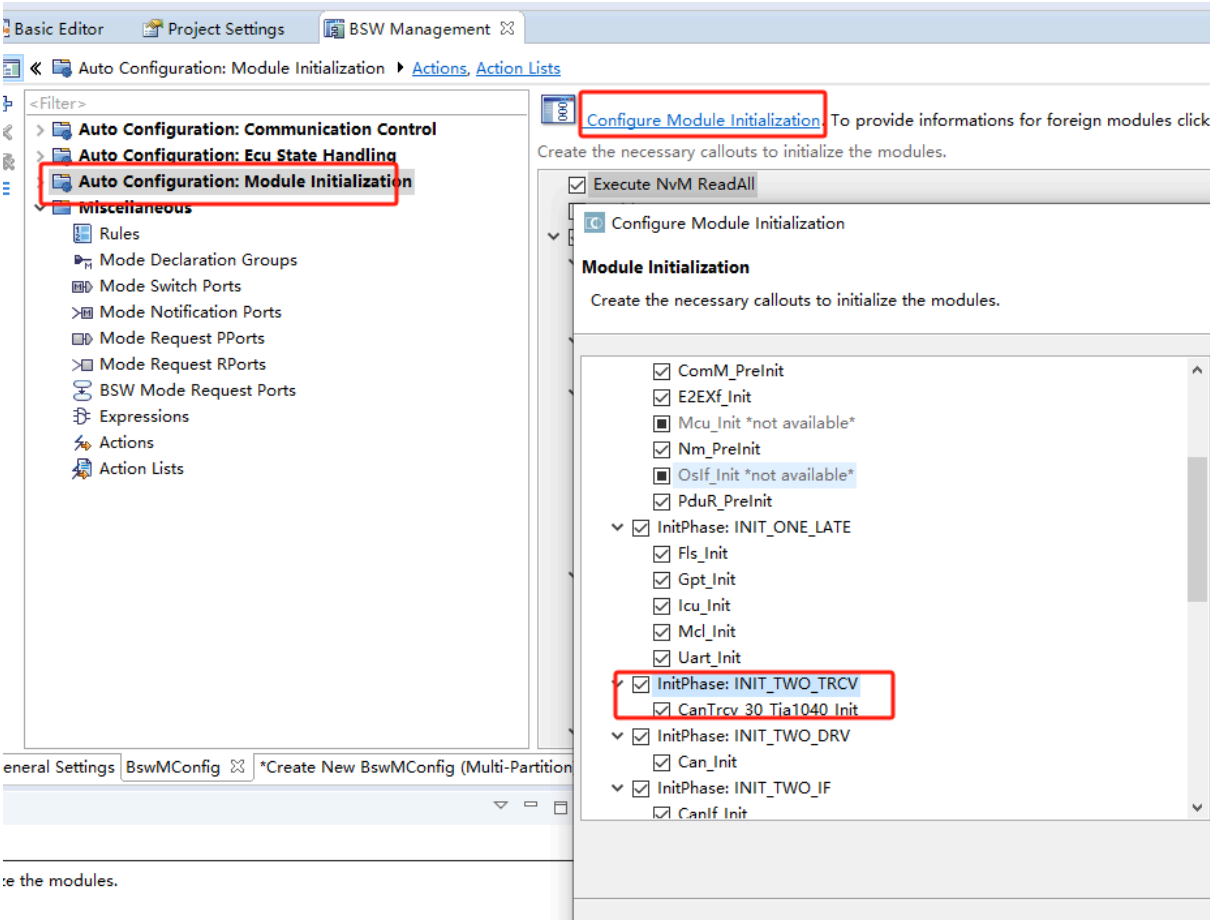

CanIf
CanIfCtrlDrvCfgs
CanIfCtrlDrvCfg_7d254554
CanIfCtrlCfgs
CT_CAN00_0e076f75
CanIfTrcvDrvCfgs
CanIfDispatchCfg
Dlc Check:
Double Hash Add [Byte]:
Double Hash Adj
Rx Search Consid
Set Pdu Reception
2 errors
Callback function / upper layer does not exist.
ID = CANIF10006
Feature "CanIfTransceiverHandling" requires the upper layer parameter "CanIfDispatchUserTrcvModeIndicationUL"
Proposed Solutions
Create parameter "CanIfDispatchUserTrcvModeIndicationUL"
Disable parameter "CanIfTransceiverHandling"
Related Items
CanIfDispatchUserTrcvModeIndicationUL
/ActiveEcuC/CanIf/CanIfPrivateCfg[0:CanIfTransceiverHandling]
Filter Ty
Tx Buffe
Extendec
Nm Osel
TTCAN:
Hand
Mapp
Fifo Sup



1.4、Rte 为Trcv的mainfunction map



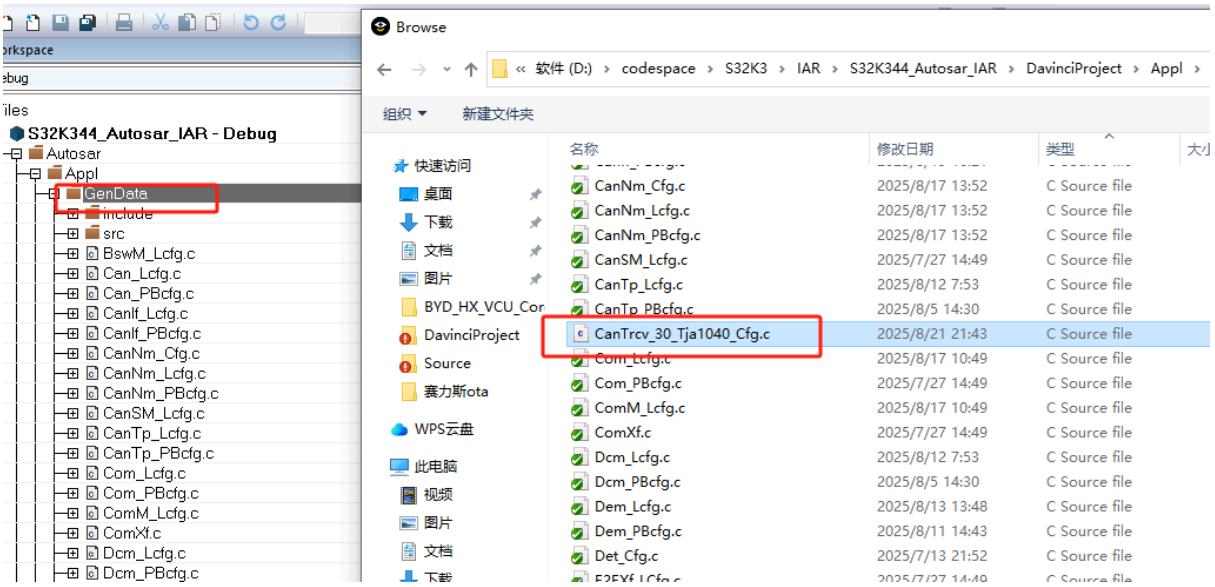
1.5、BswM 添加Trcv的Init



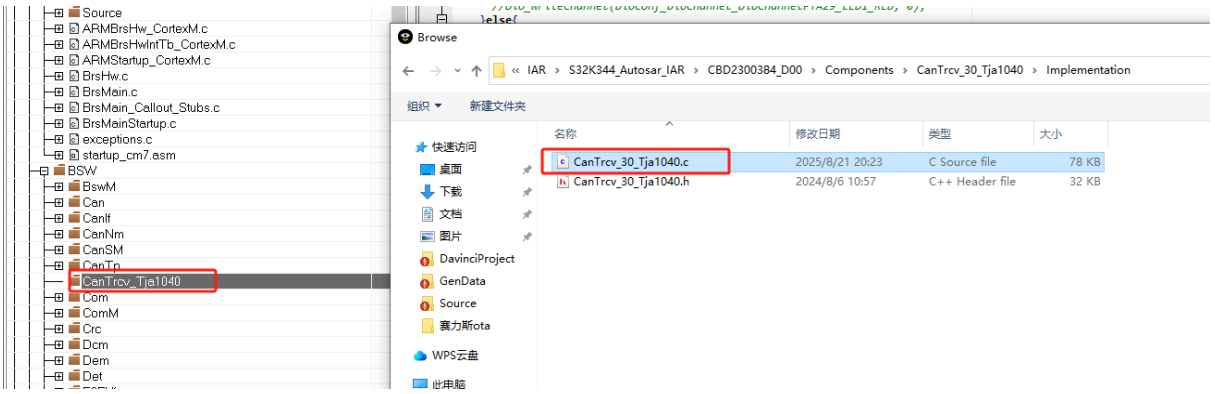
1.6、编译所有模块

二、添加CanTrcv的静态代码和生成代码

添加davinci生成的代码



添加CanTrcv的静态代码和对应的头文件路径到IAR



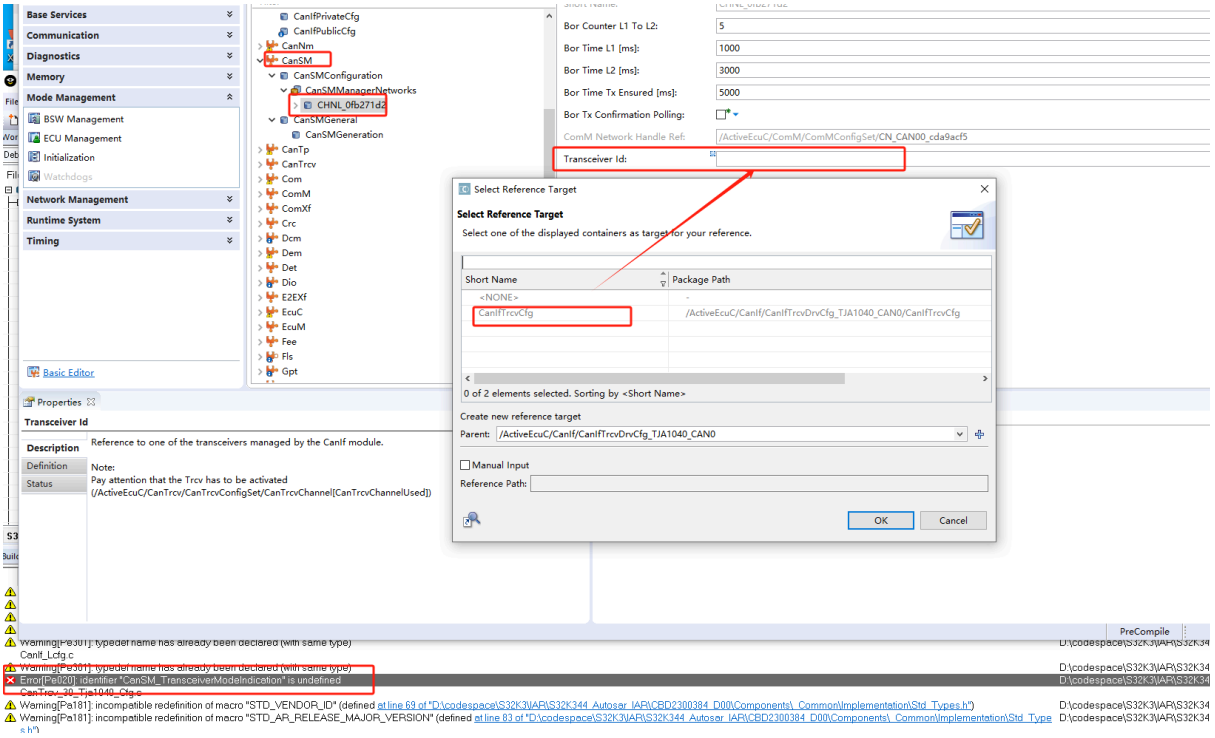
$PROJ_DIR$\CBD2300384_D00\Components\CanTrcv_30_Tja1040\Implementation

三、修改相关代码编译

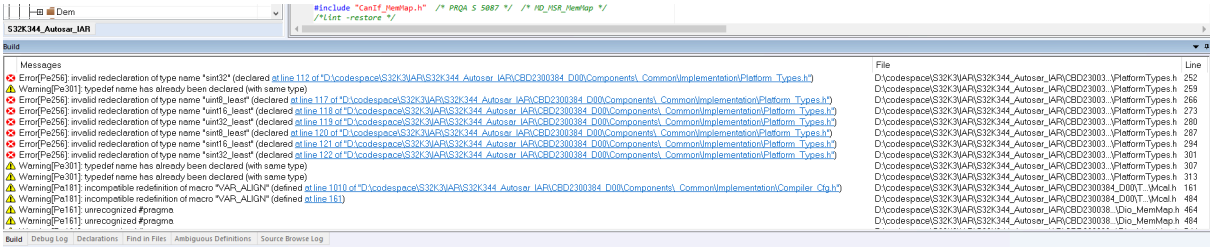
去掉LED1 dio相关的测试代码

四、编译报错修改

一个是要修改CanSM模块的



其他30个报错是跟类型重复定义，之前没出现过，这次不知道为啥出现，我们去掉mcal base 那边的头文件



Mcal_S32k › Supply › SW32K3_RTD_4.4_2.0.1 › eclipse › plugins › Base_TS_T40D34M20I1R0 › include

名称	修改日期	类型	大小
Mcl_MemMap.h	2022/7/7 17:32	C++ Header file	346 KB
Mcu_MemMap.h	2022/7/7 17:32	C++ Header file	346 KB
OsIf.h	2022/7/7 17:32	C++ Header file	8 KB
OsIf_Cfg_TypesDef.h	2022/7/7 17:32	C++ Header file	8 KB
OsIf_DeviceRegisters.h	2022/7/7 17:32	C++ Header file	5 KB
OsIf_Internal.h	2022/7/7 17:32	C++ Header file	14 KB
OsIf_Interrupts.h	2022/7/7 17:32	C++ Header file	5 KB
OsIf_Timer_Custom.h	2022/7/7 17:32	C++ Header file	5 KB
OsIf_Timer_System.h	2022/7/7 17:32	C++ Header file	5 KB
OsIf_Timer_System_Internal_Systick.h	2022/7/7 17:32	C++ Header file	7 KB
Platform_MemMap.h	2022/7/7 17:32	C++ Header file	364 KB
Platform_Types.h	2022/7/7 17:30	C++ Header file	5 KB
PlatformTypes.h	2022/7/7 17:32	C++ Header file	19 KB
Port_MemMap.h	[illegible]	[illegible]	[illegible]
Reg_eSys.h	2022/7/7 17:32	C++ Header file	8 KB
RegLockMacros.h	2022/7/7 17:30	C++ Header file	42 KB
Rte_MemMap.h	2022/7/7 17:32	C++ Header file	346 KB
Soc_Ips.h	2022/7/7 17:32	C++ Header file	10 KB
StandardTypes.h	2022/7/7 17:30	C++ Header file	8 KB
Std_Types.h	2022/7/7 17:30	C++ Header file	5 KB
Uart_MemMap.h	2022/7/7 17:32	C++ Header file	350 KB

这五个文件前面加_，相当于不用这五个文件

我们在DavinciProject\Appl 下新建两个文件，主要是对应include _Common的对应头文件

电脑 › 软件 (D:) › codespace › S32K3 › IAR › S32K344_Autosar_IAR › DavinciProject › Appl

名称	修改日期	类型	大小
ARMBrsHw_CortexM.h	2025/7/18 22:59	C++ Header file	23 KB
ARMBrsHwIntTb_CortexM.c	2025/6/17 10:46	C Source file	10 KB
ARMStartup_CortexM.c	2025/6/17 10:46	C Source file	27 KB
Brs_MemMap.h	2025/7/18 23:02	C++ Header file	11 KB
BrsHw.c	2025/7/18 23:00	C Source file	83 KB
BrsHw.h	2025/7/18 23:00	C++ Header file	19 KB
BrsHw_Ports.h	2025/6/17 10:46	C++ Header file	31 KB
BrsMain.c	2025/8/2 12:28	C Source file	48 KB
BrsMain.h	2025/7/18 23:01	C++ Header file	38 KB
BrsMain_Appl.c	2023/4/24 13:20	C Source file	12 KB
BrsMain_Appl.h	2023/4/24 13:20	C++ Header file	7 KB
BrsMain_Callout_Stubs.c	2025/7/18 23:01	C Source file	11 KB
BrsMain_Types.h	2025/6/17 10:46	C++ Header file	5 KB
BrsMainStartup.c	2025/7/18 23:01	C Source file	18 KB
BrsMainStartup.h	2025/6/17 10:46	C++ Header file	4 KB
exceptions.c	2025/7/20 22:00	C Source file	6 KB
Makefile.config.generated	2025/7/13 21:52	GENERATED 文件	5 KB
PlatformTypes.h	2025/6/17 10:51	C++ Header file	2 KB
StandardTypes.h	2025/6/17 10:51	C++ Header file	2 KB
startup_cm7.asm	2025/8/18 15:24	Assembler Source	15 KB
Vector_Table.asm	2025/7/18 23:05	Assembler Source	8 KB

添加

PlatformTypes.h ， 里面#include 的将会是_Common的 Platform_Types.h

```
PlatformTypes.h
/*****************************************************************************************************************
 *  COPYRIGHT
 *  -------------------------------------------------------------------------------------------------------------
 *  \verbatim
 *  Copyright (c) 2021 by Vector Informatik GmbH.                                          All rights reserved.
 *
 *                This software is copyright protected and proprietary to Vector Informatik GmbH.
 *                Vector Informatik GmbH grants to you only those rights as set out in the license conditions.
 *                All other rights remain with Vector Informatik GmbH.
 *  \endverbatim
 *  -------------------------------------------------------------------------------------------------------------
 *  FILE DESCRIPTION
 *  -------------------------------------------------------------------------------------------------------------*/
/**        \file  PlatformTypes.h
 *        \brief  Integration Code
 *
 *      \details  Integration Code for MCAL S32K
 *
 *  \attention    Please note:
 *                The demo and example programs only show special aspects of the software. With regard to the fact
 *                that these programs are meant for demonstration purposes only, Vector Informatik liability shall be
 *                expressly excluded in cases of ordinary negligence, to the extent admissible by law or statute.
 *****************************************************************************************************************/

#ifndef STARTAPPL_INCLUDE_PLATFORMTYPES_H_
#define STARTAPPL_INCLUDE_PLATFORMTYPES_H_

#include "Platform_Types.h"

#endif /* STARTAPPL_INCLUDE_PLATFORMTYPES_H_ */
```

StandardTypes.h , 里面#include 的将会是 *_Common的 Std*_Types.h

```
PlatformTypes.h   StandardTypes.h
/*****************************************************************************************************************
 *  COPYRIGHT
 *  -------------------------------------------------------------------------------------------------------------
 *  \verbatim
 *  Copyright (c) 2021 by Vector Informatik GmbH.
 *
 *                This software is copyright protected and proprietary to Vector Informatik
 *                Vector Informatik GmbH grants to you only those rights as set out in the
 *                All other rights remain with Vector Informatik GmbH.
 *  \endverbatim
 *  -------------------------------------------------------------------------------------------------------------
 *  FILE DESCRIPTION
 *  -------------------------------------------------------------------------------------------------------------
/**        \file  StandardTypes.h
 *        \brief  Integration Code
 *
 *      \details  Integration Code for MCAL S32K
 *
 *  \attention    Please note:
 *                The demo and example programs only show special aspects of the software.
 *                that these programs are meant for demonstration purposes only, Vector Inf
 *                expressly excluded in cases of ordinary negligence, to the extent admissi
 *****************************************************************************************************************

#ifndef STARTAPPL_INCLUDE_STANDARDTYPES_H_
#define STARTAPPL_INCLUDE_STANDARDTYPES_H_

#include "Std_Types.h"

#endif /* STARTAPPL_INCLUDE_STANDARDTYPES_H_ */
```

编译通过

```
Messages
Warning[Pe186]: pointless comparison of unsigned integer with zero
Warning[Pe186]: pointless comparison of unsigned integer with zero
Warning[Pe186]: pointless comparison of unsigned integer with zero
Warning[Pe186]: pointless comparison of unsigned integer with zero
Warning[Pa082]: undefined behavior: the order of volatile accesses is undefined in this statement
Warning[Pa082]: undefined behavior: the order of volatile accesses is undefined in this statement
Warning[Pa082]: undefined behavior: the order of volatile accesses is undefined in this statement
S32K344_Autosar_IAR.out
S32K344_Autosar_IAR.hex

Total number of errors: 0
Total number of warnings: 2635
```